Lecture 16

- Singular Values
- SVD : sing. val. decomposition
- Condition Number
- Pseudo Inverse

Let A be an $m \times n$ matrix, $A \in \mathbb{M}_{m\times n}$.

Let $G_A = A^T A$ be the Gram matrix associated with the columns of A.

Def Singular values of A, $\sigma_1, \dots, \sigma_r$, are the positive square roots of the non-zero eigenvalues of $G_A = A^T A$.

$$\boxed{\sigma_i = \sqrt{\lambda_i}}$$

- $\lambda_i \in \operatorname{spec} G_A \setminus \{0\}$
- $\sigma_i \in \operatorname{sing} A$

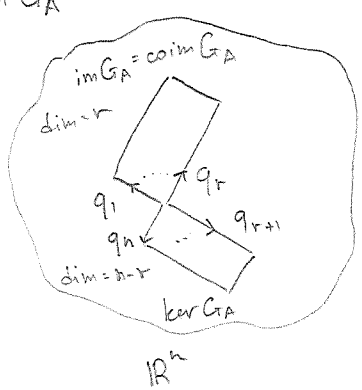
Remark: Since G_A is positive semi definite, $\lambda_i \geq 0$
$\Rightarrow \sqrt{\lambda_i}$ are well defined.

Remark: By convention, it is assumed that $\sigma_1 \geq \dots \geq \sigma_r$.

Q: What is r? (# sing. values of A) (# non-zero eigenvalues of G_A)

$G_A = A^T A$ is symmetric $\Rightarrow$ it admits an orthonormal eigenbasis of $\mathbb{R}^n$

$$\underbrace{q_1 \cdots q_r}_{\substack{\text{assoc. } \lambda \neq 0 \\ \text{basis of} \\ \text{im } G_A}} \; \underbrace{q_{r+1} \cdots q_n}_{\substack{\text{assoc. } \lambda = 0 \\ \text{basis of } \ker G_A}} \Rightarrow r = \operatorname{rank} G_A. \text{ But } \operatorname{rank} G_A = \operatorname{rank} A$$



(Recall Lect 14)

Indeed:

(a) $\ker G_A = \ker A$

- $x \in \ker A \Rightarrow Ax = 0 \Rightarrow \underbrace{A^T A}_{G_A} x = 0 \Rightarrow x \in \ker G_A$
- $x \in \ker G_A \Rightarrow A^T A x = 0 \Rightarrow x^T A^T A x = 0$
  $\Rightarrow \|Ax\| = 0 \Rightarrow Ax = 0 \Rightarrow x \in \ker A$

(b) Fund. Th of LA:

$$\left.\begin{array}{l} \dim \ker G_A = n - \operatorname{rk} G_A \\ \dim \ker A = n - \operatorname{rk} A \end{array}\right\} \Rightarrow \operatorname{rk} G_A = \operatorname{rk} A.$$

So $\boxed{r = \operatorname{rk} A}$

Q: If A is square, $A \in \mathbb{M}_{n\times n}$, then it has eigenvalues and singular values. Is there any relationship between $\operatorname{spec} A$ and $\operatorname{sing} A$?

A: In general, no.

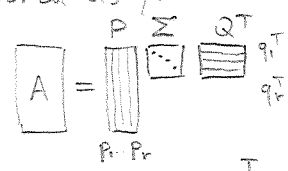
If A is symmetric $\Rightarrow \underset{\operatorname{sing} A}{\sigma_i} = \underset{\operatorname{spec} A \setminus \{0\}}{|\lambda_i|}$ (easy to check)

Singular Value Decomposition

The role of SVD for arbitrary matrices is similar to the role of spectral factorization for symmetric matrices.

Thm Let $A \in \mathbb{M}_{m\times n}$ be a matrix of rank $r > 0$. Then it can be factored as follows:

$$\boxed{A = P \cdot \Sigma \cdot Q^T}$$



- P is $m \times r$ with orthonormal columns, $P^T P = I_r$
- Σ is $r \times r$ diagonal, $\Sigma = \operatorname{diag}(\underbrace{\sigma_1, \dots, \sigma_r}_{\text{sing vals of } A})$
- Q is $n \times r$ with orthonormal columns $Q^T Q = I_r$

Remark $P = [\underbrace{p_1 \cdots p_r}_{\text{left sing. vectors of } A}]$

$Q = [\underbrace{q_1 \cdots q_r}_{\text{right sing vectors of } A}]$

Proof We want to find $P \in \mathbb{M}_{m\times r}$ and $Q \in \mathbb{M}_{n\times r}$ such that $P^TP = I$, $Q^TQ = I$, $A = P\Sigma Q^T$

Let $P = [p_1 \dots p_r]$, $Q = [q_1 \dots q_r]$

$$\Updownarrow$$

$$AQ = P\Sigma$$

Goal: to find orthonormal $p_1 \dots p_r \in \mathbb{R}^m$ and orthonormal $q_1 \dots q_r \in \mathbb{R}^n$ such that $Aq_i = \sigma_i p_i$ ← — — ith column

---

Let's take a step back. Suppose $A = P\Sigma Q^T$. What $\{q_i\}$ and $\{p_i\}$ could possibly be?

$$G_A = A^TA = Q\Sigma P^TP\Sigma Q^T = Q\underbrace{\Sigma^2}_{\text{diag}(\operatorname{spec} G_A \setminus \{0\})}Q^T$$

$$G_AQ = Q\Sigma^2 \quad \text{or} \quad G_Aq_i = \sigma_i^2 q_i$$

$\Rightarrow$ q_i is an orthonormal eigenvector of G_A associated with $\sigma_i^2 \in \operatorname{spec} G_A \setminus \{0\}$. This tells us how to proceed.

Since $G_A = A^TA$ is symmetric $\Rightarrow \exists$ orthonormal eigenbasis of $\mathbb{R}^n$

Let $q_1 \dots q_r$ be the orthonormal eigenvectors of G_A associated with non-zero eigenvalues. And let $p_i = \dfrac{Aq_i}{\sigma_i}$

All we need to do now is to check that

(a) p_i have unit norm:

$$\|p_i\|^2 = p_i^Tp_i = \frac{1}{\sigma_i^2}(Aq_i)^T(Aq_i) = \frac{q_i^T\overbrace{A^TA}^{G_A}q_i}{\sigma_i^2} = \frac{\lambda_i\|q_i\|^2}{\sigma_i^2} = \underline{1}$$

(b) p_i are orthogonal

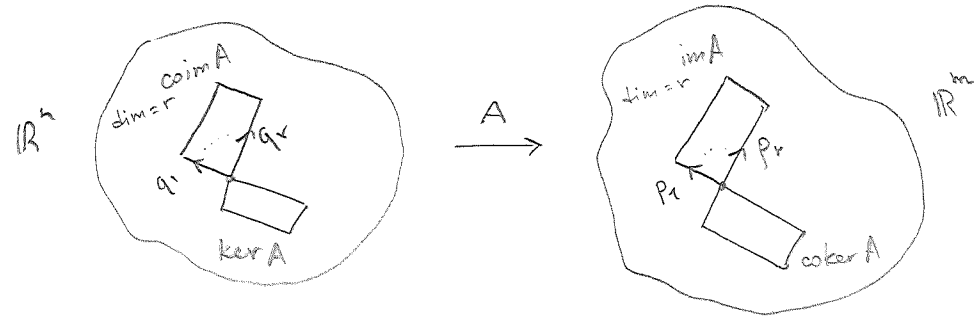
$$\langle p_i, p_j\rangle = \frac{(Aq_i)^T(Aq_j)}{\sigma_i\sigma_j} = \frac{q_i^TG_Aq_j}{\sigma_i\sigma_j} = \frac{\lambda_j\overbrace{q_i^Tq_j}^{0}}{\sigma_i\sigma_j} = 0$$

$\square$

Remark: $\{p_i\}$ are orthonormal eigenvectors of AA^T associated with non-zero eigenvalues.

Singular vectors $\{q_i\}$ and $\{p_i\}$ have nice geometric interpretation

Fund Theorem:



- $p_i = \dfrac{Aq_i}{\sigma_i} = A\left(\dfrac{q_i}{\sigma_i}\right) \in \operatorname{im} A$, $\dim \operatorname{im} A = r$

  $\Rightarrow$ $p_1 \dots p_r$ form an orthonormal basis of $\operatorname{im} A = \operatorname{span}(a_1 \dots a_n)$

- If $A = P\Sigma Q^T \Rightarrow A^T = Q\Sigma P^T$ ← This is SVD of A^T since $\operatorname{sing} A^T = \operatorname{sing} A$.

$\Rightarrow$ $q_1 \dots q_r$ form an orthonorm. basis of $\operatorname{im} A^T = \operatorname{coim} A = \operatorname{span}(a^1 \dots a^m)$

This follows from the fact that

$$\boxed{\operatorname{spec} A^TA \setminus \{0\} = \operatorname{spec} AA^T \setminus \{0\}}$$

as sets of points.

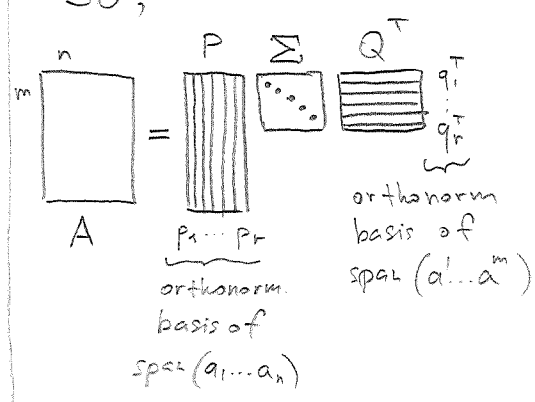
Indeed: $\lambda \in \operatorname{spec} A^TA$, $\lambda \neq 0$

$\Rightarrow \exists v \neq 0 : A^TAv = \lambda v$

$\Rightarrow AA^T(Av) = \lambda(Av)$

$\Rightarrow \lambda \in \operatorname{spec} AA^T$ since $Av \neq 0$

So,



Condition Number

Let $A \in \mathbb{M}_{n\times n}$ and $r = \operatorname{rank} A$

- If $r < n \Rightarrow A$ is singular
- If $r = n \Rightarrow A$ is nonsingular and $Ax = b$ has unique solution.

But if A is "close" to being singular $\Rightarrow$ solving $Ax = b$ can be difficult.

Condition number $\varkappa(A)$ quantifies "closeness" of A to singularity

Def Let $A \in \mathbb{M}_{n\times n}$ be nonsingular.

Then $\varkappa(A) = \dfrac{\sigma_1}{\sigma_n}$ $\leftarrow$ largest sing val, $\leftarrow$ smallest sing val.

If A is sing $\Rightarrow \varkappa = \infty$.

In general, the larger $\varkappa$ is, the closer A to being singular, the more difficult to solve $Ax = b$.

If $\varkappa(A)$ is large $\Rightarrow A$ is ill-conditioned

$\varkappa > \dfrac{1}{\text{machine precision}}$ $(\approx 10^7)$

Pseudoinverse (aka Moore-Penrose inverse)

~1920s ~1950s

The SVD allows to generalize the notion of inverse matrix

Def Let $A \in \mathbb{M}_{m\times n}$ and $A = P\Sigma Q^T$ be its SVD.

The pseudoinverse $\boxed{A^+ = Q\Sigma^{-1}P^T}$

(1) If A is nonsingular $\Rightarrow A^+ = A^{-1}$

Indeed:

$$A^{-1} = (P\Sigma Q^T)^{-1} = (Q^T)^{-1}\Sigma^{-1}P^{-1} = Q\Sigma^{-1}P^T = A^+$$

(2) If $\ker A = \{0\}$ ($\Leftrightarrow \operatorname{rk} A = n$) $\Rightarrow$ to compute A^+ we don't need the SVD of A since $A^+ = (A^TA)^{-1}A^T$

Indeed: $(A^TA)^{-1}A^T = (Q\Sigma P^TP\Sigma Q^T)^{-1}Q\Sigma P^T$

$$= (Q\Sigma^2Q^T)^{-1}Q\Sigma P^T \underset{\uparrow\, Q\in\mathbb{M}_{n\times n}\text{ orthogonal}}{=} Q\Sigma^{-2}Q^TQ\Sigma P^T = Q\Sigma^{-1}P^T = A^+$$

Q: Do $A^+ = (A^TA)^{-1}A^T$ look familiar?

Yes! If $\ker A = \{0\} \Rightarrow x^* = (A^TA)^{-1}A^Tb$ is the (unique) least squares solution of $Ax = b$.

Thm Consider $Ax = b$ and let $x^* = A^+b$

- If $\ker A = \{0\} \Rightarrow x^*$ is the l.s.s. of $Ax = b$.
- If $\ker A \neq \{0\} \Rightarrow \|Ax - b\| \to \min$ has ∞-many solutions
  x^* is the one with smallest norm
  general solution: $x^* + \ker A$.